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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
VII Semester B. E. Regular Examinations March/April-2025
Computer Science & Engineering

PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

- Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
- Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A			M	BT	CO
1	1.1	The fetch and execution cycles are interleaved with the help of _____.	01	2	1
	1.2	The periods of time when the unit is idle is called as _____.	01	1	2
	1.3	Define register pressure which is a scenario that emerges as a limitation of unrolling.	02	2	1
	1.4	How are branch conditions specified in <i>RISC</i> architectures?	02	2	2
	1.5	List the orderings that sequential consistency needs to maintain.	02	2	1
	1.6	The following code sequence lays out into how many convoys? Indicate them. <i>LV V1,Rx ; load vector X</i> <i>MULVS.D V2,V1,F0; vector – scalar multiply</i> <i>LV V3,Ry; load vector Y</i> <i>ADDVV.D V4,V2,V3; add two vectors</i> <i>SV V4,Ry; store the sum</i>	02	4	3
	1.7	With multiple threads, shared memory in <i>CUDA</i> is accessible to _____.	01	3	2
	1.8	Parallel portions of an application are executed on the device are referred to as _____.	01	2	1
	1.9	In the following function call, a message is sent to which process? <i>MPI_Send(message ,4,MPI_CHAR,5,tag,MPI_COMM_WORLD)</i>	02	3	3
	1.10	Name two additional datatypes that <i>MPI</i> allows which are not part of the C language.	02	3	3
	1.11	The number and size of tasks into which a problem is decomposed determines the _____ of the decomposition.	01	2	2
	1.12	In OpenACC, to implement overlapping of computation with data transfer/PCIe copies, a technique _____ could be used that breaks the given problem to smaller independent chunks.	01	2	3
	1.13	OpenACC does not provide any lock mechanism except for _____ operations to prevent race conditions.	01	3	3
	1.14	What is the core component of a <i>TPU</i> responsible for high-speed matrix multiplication?	01	2	3

PART-B

2	a	Suppose the following measurements have been made: - Frequency of <i>FP</i> operations = 25% - Average <i>CPI</i> of <i>FP</i> operations = 4.0 - Average <i>CPI</i> of other instructions= 1.33 - Frequency of <i>FPSQR</i> = 2% <i>CPI</i> of <i>FPSQR</i> = 20 Assume that the two design alternatives are to decrease the <i>CPI</i> of <i>FPSQR</i> to 2 or to decrease the average <i>CPI</i> of all <i>FP</i> operations to 2.5. Compare these two design alternatives using the processor performance equation.	05	3	2
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	b	Consider a design consideration for enhancement to the processor of a server. The new <i>CPU</i> is 10 times faster. <i>I/O</i> bound server, which therefore spends 60% time waiting for <i>I/O</i> . What is the overall speedup achieved?	04	3	4
	c	Discuss the three different types of dependences. Illustrate how it constraints the amount of parallelism in programs.	07	3	2
3	a	Explain snooping cache coherence protocol with state transition diagram based on request generated from <i>CPU</i> .	08	2	2
	b	Draw and explain different multiprocessor architectures. OR	08	2	2
4	a	Explain how coherence problem is solved by directory based write invalidate protocol with a neat transition diagram for the directory structure.	08	2	2
	b	What is multiprocessor cache coherence? Differentiate between coherence and consistency. Discuss the basic schemes for enforcing coherence.	08	2	2
5	a	Discuss the major innovations introduced by Fermi to <i>GPU</i> architecture.	08	2	4
	b	Show the code of <i>VMIPS</i> for the <i>DAXPY</i> loop given below. Assume that the starting addresses of <i>X</i> and <i>Y</i> are in <i>R_x</i> and <i>R_y</i> , respectively. $Y = a * X + Y$	04	4	4
	c	Write <i>CUDA</i> code snippet for the <i>DAXPY</i> loop given below, to launch <i>n</i> threads, one per vector element, with 256 <i>CUDA</i> threads per Thread block in multithreaded <i>SIMD</i> processor. $Y = a * X + Y$ OR	04	4	3
6	a	Draw the basic structure of vector architecture for <i>VMIPS</i> and explain its components.	08	2	2
	b	Write a simple function in <i>CUDA</i> for matrix multiplication with only host code, discuss the salient aspects of <i>CUDA</i> device to execute the same. Revise the function that moves matrix multiplication to a device.	08	4	3
7	a	Discuss the characteristics of inter-task interactions highlighting the salient aspects of each.	08	2	2
	b	Differentiate between blocking and non-blocking message passing operations. OR	08	2	3
8	a	Describe the static mapping techniques used in parallel algorithms.	09	2	2
	b	List and explain the set of functions employed for performing commonly used collective communication operations in <i>MPI</i> .	07	2	3
9	a	Illustrate with a pseudo code (with appropriate calling sequence of OpenACC directives), implementation of pipelining in which the data transfers and computations are blocked.	10	4	3
	b	Discuss the various forms of expressing parallelism in OpenACC. OR	06	3	3
10	a	What are the different levels of parallelism that can be expressed by a OpenACC programmer? Appropriately highlight the three rules of offload mode programming. Discuss the challenges faced while locks are employed to protect a critical region and propose a solution to the above discussion.	10	3	3
	b	Explain the working of <i>TPU</i> . Differentiate <i>TPU</i> verses <i>GPU</i> .	06	3	4